

Personal Data Protection in Hubgate

PII Masking — Client Guide

Keeper Solutions

1. Introduction

Hubgate includes a built-in **personal data protection layer** that automatically shields sensitive information before it is sent to an external AI provider.

When you work with an AI assistant, your messages are processed by a third-party AI model (such as those operated by OpenAI, Anthropic, or Google). Without protection, any personal data contained in a message — a name, an email address, a national identification number, a payment card — would be transmitted to that provider in its original form.

Hubgate's PII masking feature prevents this. It detects personal data in your message, replaces each value with a neutral placeholder before the message leaves your environment, and restores the original values in the response shown to you. The protection is automatic and requires no special action on your part.

This approach supports your organisation's privacy and regulatory obligations, including GDPR and NIS2, while preserving the full usefulness of the AI assistant.

2. How it works

The protection operates in two directions and is invisible in normal use:

- **Before the message is sent**, Hubgate scans it, identifies any personal data, and replaces each value with a placeholder. The AI model therefore only ever receives masked text.
- **When the response returns**, Hubgate restores the original values in place of the placeholders, so the reply you read contains your real data.

In short: the external AI provider never sees the original personal data, while you continue to see and work with your real information exactly as expected.

A simple illustration:

You write → Personal data detected and masked → Sent to the AI model

AI responds → Original values restored → Shown to you

3. How personal data is detected

Hubgate identifies personal data using a combination of complementary detection methods, so that both free-form text and strictly formatted identifiers are recognised reliably:

- **Microsoft Presidio** — an established open-source data-protection framework that provides the core detection and masking engine. It uses the **spaCy** natural-language library to understand the structure of the text.
- **GLiNER** — a modern neural model used to recognise context-dependent information such as personal names, addresses, usernames, IP addresses, and secrets (passwords, API keys, and tokens).
- **Custom validated recognisers** — purpose-built rules developed by Keeper Solutions for national identifiers and bank account numbers (for example, Croatian OIB, IBANs, or UK National Insurance numbers). These additionally verify each value's internal check digits, so that only genuine identifiers are masked and accidental sequences are left untouched.

Together these methods cover names, contact details, financial data, national identifiers, and technical credentials across both Croatian and English text.

4. What is protected

Hubgate recognises a broad range of personal and sensitive data, grouped into clear categories.

Category	Examples
Names and identity	Personal names, postal addresses
National identifiers	Croatian OIB and JMBG, Irish PPS number, Romanian CNP, UK National Insurance

Category	Examples
	number and tax reference, UK NHS number, US Social Security and Employer ID numbers
Financial information	Payment card numbers, IBAN bank account numbers (Croatian, Irish, Romanian, UK, and other European formats)
Contact details	Email addresses, telephone numbers
Credentials and technical data	Username and login handles, IP addresses, passwords, API keys, and access tokens

4.1 Complete list of recognised data

The table below lists every type of data Hubgate can detect, the detection method it comes from, and an example. The placeholder labels are illustrative.

Detected by	Data type	Example input	Replaced with
GLiNER	Personal name	James Whitfield	[PERSON_1]
GLiNER	Postal address	12 Northgate St, Dublin	[ADDRESS_1]
GLiNER	Username / login handle	@aoife.b	[USERNAME_1]
GLiNER	IP address	192.168.1.10	[IP_ADDRESS_1]
GLiNER	Secret (password, API key, token)	sk-live-9f2a8c0b4d1e	[SECRET_1]
Presidio	Email address	james@northgate.co.uk	[EMAIL_1]
Presidio	Telephone number	+1 202 456 1111	[PHONE_1]
Presidio	Payment card number	4111 1111 1111 1111	[CREDIT_CARD_1]
Presidio	UK NHS number	943 476 5919	[UK_NHS_1]

Detected by	Data type	Example input	Replaced with
Presidio	IBAN (other countries, e.g. German)	DE89 3704 0044 0532 0130 00	[IBAN_1]
Custom	Croatian OIB	12345678903	[HR_OIB_1]
Custom	Croatian JMBG	0101990330021	[HR_JMBG_1]
Custom	Croatian IBAN	HR12100100518630 00160	[HR_IBAN_1]
Custom	Irish PPS number	1234567FA	[IE_PPSN_1]
Custom	Irish IBAN	IE29AIBK93115212 345678	[IE_IBAN_1]
Custom	Romanian CNP	1960101221144	[RO_CNP_1]
Custom	Romanian IBAN	R049AAAA1B310075 93840000	[RO_IBAN_1]
Custom	UK National Insurance number	AB123456C	[UK_NINO_1]
Custom	UK tax reference (UTR)	UTR 1234567890	[UK_UTR_1]
Custom	UK IBAN	GB29NWBK60161331 926819	[GB_IBAN_1]
Custom	US Social Security number	123-45-6789	[US_SSN_1]
Custom	US Employer ID number	EIN 12-3456789	[US_EIN_1]

Note. The examples illustrate format only. Structured identifiers (national IDs, IBANs, payment cards) are validated for correctness before they are masked. The number at the end of each placeholder increases for each distinct value within a conversation.

Personal data is detected reliably across both Croatian and English text.

5. Examples

The examples below show how a message changes on its way to the AI model.

Example 1 — name, email, and national identifier

Stage	Text
You write	My name is James Whitfield, reach me at james.whitfield@northgate.co.uk, NI number AB123456C.
Sent to the AI model	My name is [PERSON_1], reach me at [EMAIL_1], NI number [UK_NINO_1].
Shown to you	The original values, restored automatically. The placeholders are never shown.

Example 2 — a customer support message

Stage	Text
You write	Customer Aoife Brennan, card 4111 1111 1111 1111, IBAN DE89 3704 0044 0532 0130 00, login aoife.b – please process a refund.
Sent to the AI model	Customer [PERSON_1], card [CREDIT_CARD_1], IBAN [IBAN_1], login [USERNAME_1] – please process a refund.

Example 3 — consistency within a conversation

Within a single conversation, the same value is always represented by the same placeholder. This keeps the dialogue coherent for the AI model without ever revealing the underlying data.

Turn	Text
Earlier message	"Email Aoife Brennan the report." becomes [PERSON_1]
Later message	"Did Aoife reply?" uses the same [PERSON_1]

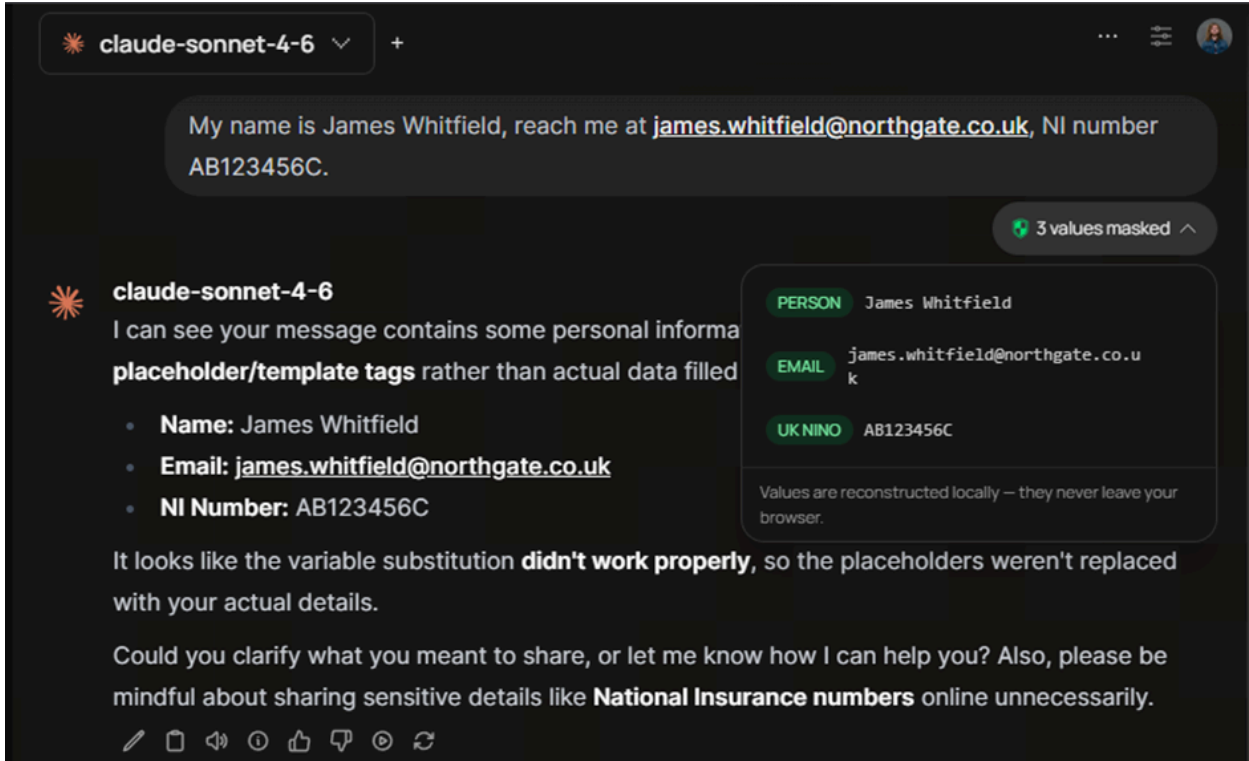


Figure 1. Personal data masking in a Hubgate conversation: the original message, and the assistant's reply with the real values restored.

6. Using PII masking in Hubgate

Personal data protection is **enabled by default**. No configuration is required to benefit from it. For situations where you need finer control, Hubgate provides two settings.

6.1 Global setting

In your **Privacy settings**, a single control enables or disables PII masking for all of your conversations. This is the default protection level for everything you do in Hubgate. The same screen also lists the categories of data that are protected, so you can see at a glance what is covered.

By default this setting is **on**.

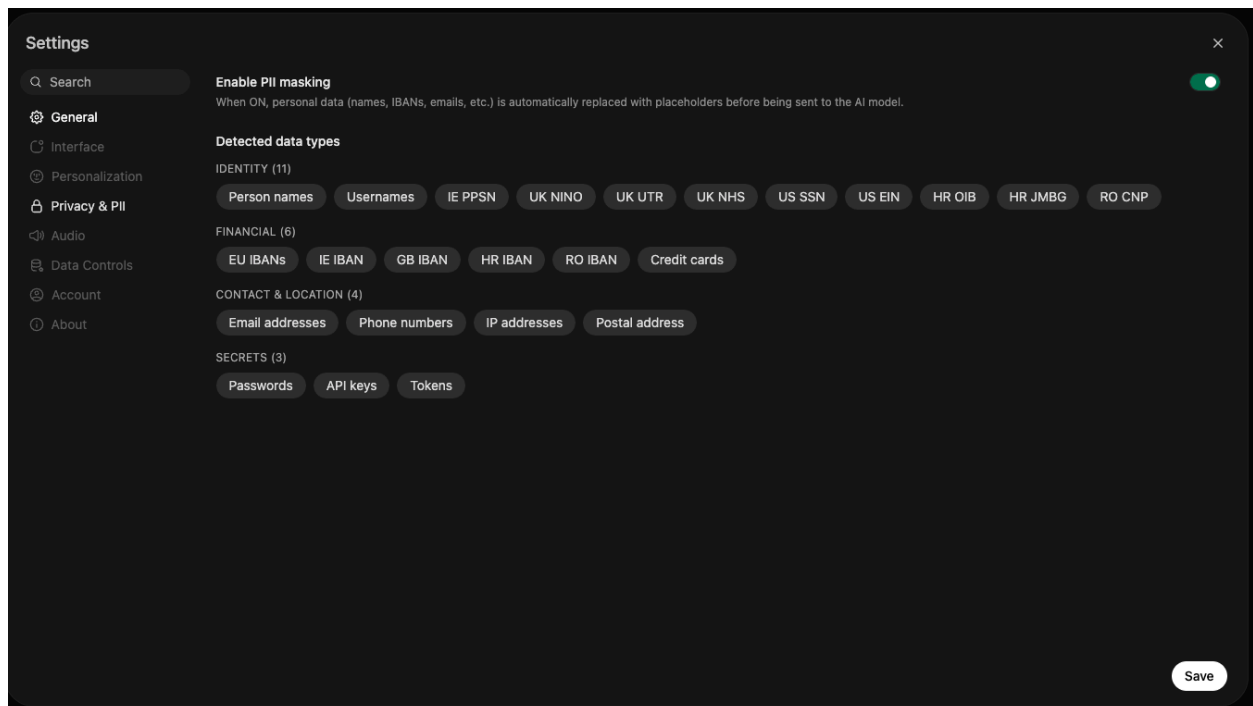


Figure 2. Privacy settings: the PII masking toggle, the Save action, and the list of protected data categories.

6.2 Per-conversation control

Inside any conversation, the message bar includes a toggle that enables or disables masking **for that conversation only**, without changing your global setting. This is useful when a particular conversation needs a different level of protection from your normal default.

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Figure 3. The per-conversation PII toggle in the message bar.

6.3 What enabling and disabling means

Setting	Result
Enabled	Every message is protected. Personal data is masked before it reaches the AI model and restored in the reply.
Disabled	Messages are sent to the AI model in their original form, without masking.

Example — protection enabled

Stage	Text
You send	Call Mr. Dawson at +1 202 456 1111 about the invoice for card 4111 1111 1111 1111.
Sent to the AI model	Call [PERSON_1] at [PHONE_1] about the invoice for card [CREDIT_CARD_1].
Shown to you	The original values, fully restored in the reply.

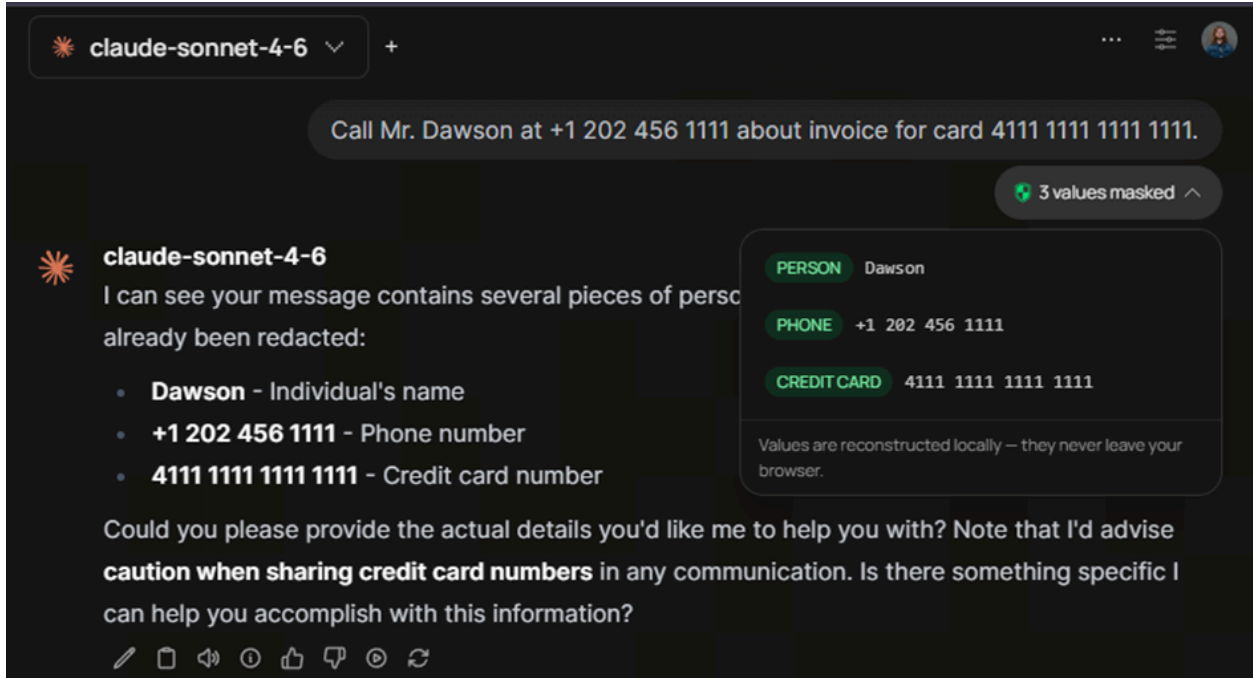


Figure 4. A conversation with protection enabled: personal data is masked on the way to the AI model and restored in the reply shown to you.

Example — protection disabled

Stage	Text
You send	Call Mr. Dawson at +1 202 456 1111 about the invoice for card 4111 1111 1111 1111.
Sent to the AI model	Call Mr. Dawson at +1 202 456 1111 about the invoice for card 4111 1111 1111 1111.

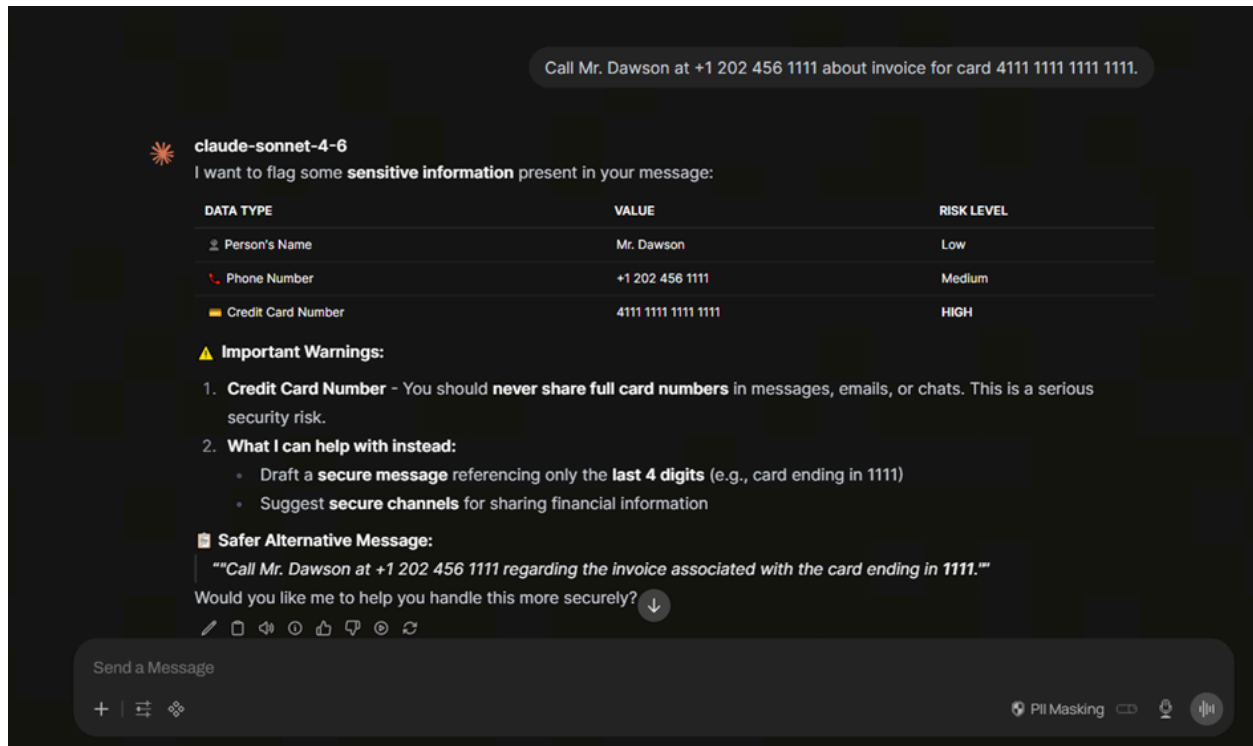


Figure 5. A conversation with protection disabled: the message reaches the AI model with its original values.

Important. When protection is disabled, personal data is transmitted to the external AI provider in its original form. This option should be used only when you are certain that the message contains no sensitive information.

7. How protected data is stored

To restore your original values in the reply, Hubgate must briefly remember which placeholder corresponds to which value. This link is recorded in a secure database, with the following safeguards:

- **Stored in encrypted form.** The original values are saved encrypted, so they are not readable in storage.
- **Separated per conversation.** Each conversation has its own isolated set of mappings; the same value used in two different conversations is never linked.
- **Automatic expiry.** Mappings are kept only as long as needed and are removed automatically after a limited period.

- **Used only to restore your data.** This stored information serves a single purpose — putting your original values back into the AI's reply. It is never sent to the external AI provider.

chat_id	entity_type	original_value	lookup_hash	placeholder	counter_index	created_at	expires_at
b3e4a64f-c3...	PHONE	ENC1:AQAAAAAG5FS8AqG5AdTJSolIK3quXL9Fw3pbsxgnGULdKa...	UJNAKJL+8&[VT]Qq[NUL]Sz...	[PHONE_1]	1	2026-06-24T11:29:2...	2026-06-25T11:30...
b3e4a64f-c3...	CREDIT_CARD	ENC1:AQAAAAHP3k/nZC5+R+SsIQ2LcFVWxc2KRpUGx4POUivn...	0&w[EM][ENQ]cGqY\DC3]...	[CREDIT_CARD_1]	1	2026-06-24T11:29:2...	2026-06-25T11:30...
b3e4a64f-c3...	PERSON	ENC1:AQAAAAFDgWmfzv9MzSYsBwBX5MOvYsgKaWMjx6CGB...	A^*4[DC3]ITIo%[SO]_1q3 Ca...	[PERSON_1]	1	2026-06-24T11:29:2...	2026-06-25T11:30...
ea7bbadd-0...	EMAIL	ENC1:AQAAAAESDvxxVpr2M6wc4/a3hs0R2r5Ux/LBLjodiU6/soZ...	AQ[ACK]~+ft~>-<[VT][GS]ubt...	[EMAIL_1]	1	2026-06-24T11:20:5...	2026-06-25T11:25...
ea7bbadd-0...	CREDIT_CARD	ENC1:AQAAAAAGozIXN8&rfo8dz0/co1eBPAAxwF8tpfd5fJgF...	*tPtz[FS]>w6Cr[SO][DC4]...	[CREDIT_CARD_1]	1	2026-06-24T11:22:2...	2026-06-25T11:25...
ea7bbadd-0...	IBAN_CODE	ENC1:AQAAAAFYN5wm6Mpu9eaS9OzWXinw9oBRis7WmzQVD...	F#[US]b[ESC]ZSX[DC4]Tb4...	[IBAN_CODE_1]	1	2026-06-24T11:22:2...	2026-06-25T11:25...
ea7bbadd-0...	PERSON	ENC1:AQAAAAAGWx7m5sUJWsWbK4cd7L4mhIV46SDoK08BD/D8...	[GS][Y.M][SO]u[FS]Cz-[DEL...	[PERSON_3]	3	2026-06-24T11:22:2...	2026-06-25T11:25...
ea7bbadd-0...	PERSON	ENC1:AQAAAAHVkuSo4wYpCIeun0SroOLyPiY1Fsqtzyp8hgqH8...	[DC4] <Rvu[EM]g9v[ETX]k...	[PERSON_1]	1	2026-06-24T11:20:5...	2026-06-25T11:25...
ea7bbadd-0...	UK_NINO	ENC1:AQAAAAECpFTwgGoPQ5OWUI3tyMYdthOmU6oac+RS4...	vcg[PV]BS] U ASs ▲	[UK_NINO_1]	1	2026-06-24T11:20:5...	2026-06-25T11:25...
ea7bbadd-0...	PERSON	ENC1:AQAAAAEVL7uIRsYTPwdFDzURQeabdsF/xbgjnZw0Btwm...	.d2w@[SUB][SUB]#M[STX]...	[PERSON_5]	5	2026-06-24T11:23:2...	2026-06-25T11:25...
ea7bbadd-0...	PERSON	ENC1:AQAAAAAGICPnn7PmxTyRUygTm6B9ZgyKHpdX56XdigjvE...	W[GS][ETB]™ IE9T[SOH]S[0...	[PERSON_10]	10	2026-06-24T11:24:3...	2026-06-25T11:25...

Figure 6. The secure, encrypted store that holds the placeholder-to-value mappings for each conversation.

8. File attachments (available soon)

Note. The capabilities described in this section are being introduced and will be available in an upcoming release.

Hubgate will extend the same personal data protection to files that you attach to a message. The protection works as follows:

- When you attach a file, Hubgate does not send the whole file to the AI model. Instead, it reads the file's content and works with only the parts that are relevant to your question.
- Before any part of the file is passed to the AI model, that content is masked in exactly the same way as your messages, using the same protection applied throughout the conversation.
- If, for any reason, the content cannot be masked, the request is stopped and **nothing unmasked is sent** to the AI model. You are informed that the content could not be protected.

When you receive an answer based on an attached file, Hubgate will show:

- **Sources**, presented as numbered references, so you can see which parts of the file the answer is based on.
- **An overview of the protected data**, indicating which types of sensitive information were found in the file and where, so you have full visibility of what was masked.

The actual sensitive values are never transmitted unnecessarily between the interface and the underlying service.

9. Data protection assurances

The PII masking feature is designed around a small set of clear principles:

- **No personal data reaches the AI provider unmasked.** When protection is enabled, original values are replaced before the message or file content leaves your environment.
 - **Restored values are always accurate.** The original information shown in a reply is taken only from the protected mapping for that conversation; it is never guessed or generated by the AI model.
 - **Consistency within a conversation.** A given value is represented by the same placeholder throughout a conversation, keeping the dialogue coherent without exposing the data.
 - **Protected storage with automatic expiry.** The information needed to restore your data is held in encrypted form and is automatically removed after a limited period.
 - **Safe by default.** If protection cannot be applied for any reason, the request is stopped rather than sending unprotected data. The default behaviour always favours privacy.
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10. Summary

Hubgate's PII masking provides automatic, two-way protection for personal data when you use AI assistants. It combines a proven detection framework, a modern neural model, and custom validated rules to recognise a wide range of personal and sensitive information. The feature works without effort on your part, is enabled by default, and gives you straightforward control at both the global and per-conversation level — delivering the full benefit of an AI assistant while keeping your, and your customers', personal data private.
